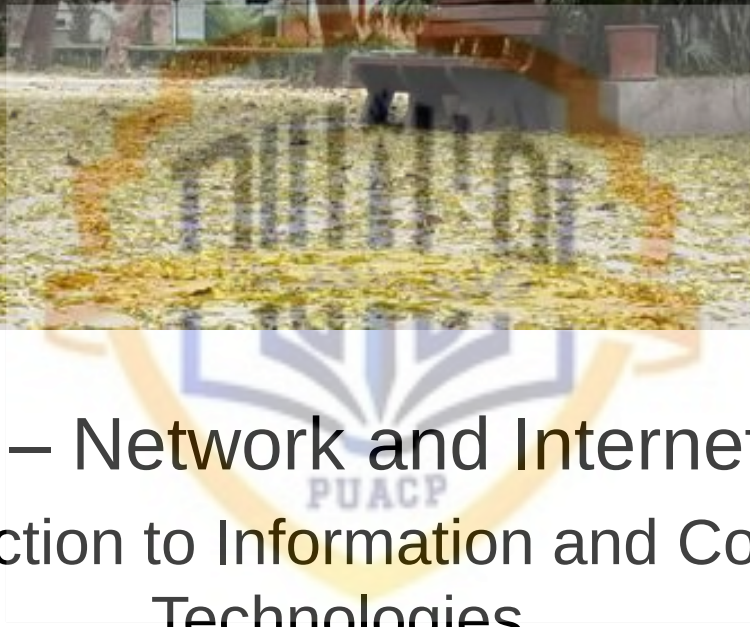




Punjab University College
of
Information Technology
“PUCIT”



Chapter 09 – Network and Internet Security

GE-161 Introduction to Information and Communication Technologies

Prof. Dr. Syed Waqar ul Qounain
Department of Information Technology
University of the Punjab, Lahore

Learning Objectives

1. Explain why computer users should be concerned about network and Internet security.
2. List several examples of unauthorized access and unauthorized use.
3. Explain several ways to protect against unauthorized access and unauthorized use, including access control systems, firewalls, and encryption.
4. Provide several examples of computer sabotage.
5. List how individuals and businesses can protect against computer sabotage.
6. Discuss online theft, identity theft, spoofing, phishing, and other types of dot cons.

Learning Objectives

7. Detail steps an individual can take to protect against online theft, identity theft, spoofing, phishing, and other types of dot cons.
8. Identify personal safety risks associated with Internet use.
9. List steps individuals can take to safeguard their personal safety when using the Internet.
10. Discuss the current state of network and Internet security legislation.

Overview

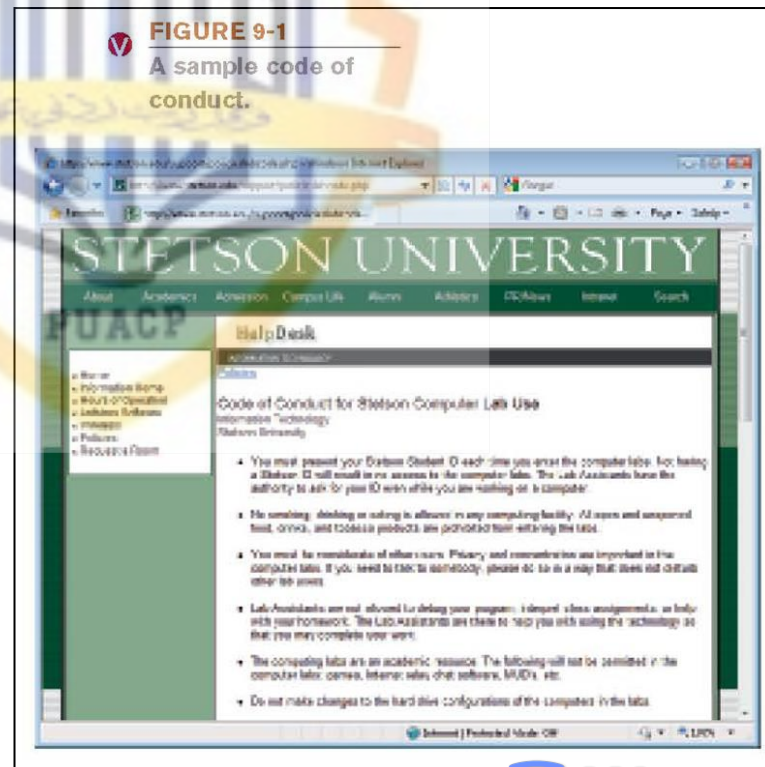
- This chapter covers:
 - Security concerns stemming from the use of computer networks
 - Safeguards and precautions that can be taken to reduce the risk of problems related to these security concerns
 - Personal safety issues related to the Internet
 - Ways to protect against personal safety issues
 - Legislation related to network and Internet security

Why Be Concerned about Network and Internet Security?

- Security concerns related to computer networks and the Internet abound
- Computer crime (cybercrime): Any illegal act involving a computer, including:
 - Theft of financial assets
 - Manipulating data for personal advantage
 - Act of sabotage (releasing a computer virus, shutting down a Web server)
- All computer users should be aware of security concerns and the precautions that can be taken

Unauthorized Access and Unauthorized Use

- Unauthorized access: Gaining access to a computer, network, file, or other resource without permission
- Unauthorized use: Using a computer resource for unapproved activities
- Both can be committed by insiders and outsiders
- Codes of conduct: Used to specify rules for behavior, typically by a business or school



Unauthorized Access and Unauthorized Use

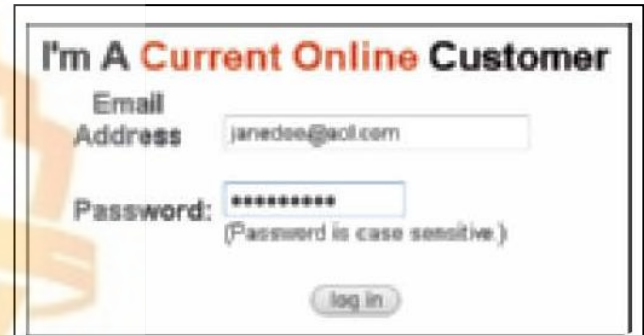
- Hacking: Using a computer to break into another computer system
 - A serious threat for individuals, businesses, and the country (national security)
 - Often performed via wireless networks today
 - Many wireless networks are left unsecured
- War driving: Driving around an area to find a Wi-Fi network to access and use without authorization
- Wi-Fi piggybacking: Accessing an unsecured Wi-Fi network from your current location without authorization
- Interception of communications: Messages, files, logon information etc. can be intercepted in not secured

Protecting Against Unauthorized Access and Use

- Access control systems: Used to control access to:
 - Facilities
 - Computer networks
 - Databases
 - Web site accounts
- Can be:
 - Identification systems: Verify that the person trying to access the facility or system is an authorized user
 - Authentication systems: Determine if the person is who he or she claims to be

Access Control Systems

- Possessed knowledge access systems: Use information that only an individual should know
 - Usernames
 - Passwords
 - Should be strong passwords and changed frequently
 - Tokens can generate passwords
 - Disadvantages: Can be forgotten and used by someone else
 - Cognitive authentication systems: Use information the individual knows (birthplace, pet names, etc.)
 - Used in many password recovery systems



The screenshot shows a login interface with the heading "I'm A Current Online Customer". It includes an "Email Address" field with the value "janedee@aol.com", a "Password" field with masked characters "*****", and a note "(Password is case sensitive.)". A "log in" button is located at the bottom right of the form.



FIGURE 9-3

Passwords.

Passwords are used to log on to computers, networks, Web sites, and other computing resources.

Possessed Knowledge Systems

FIGURE 9-4
Strategies for
creating good
passwords.

PASSWORD STRATEGIES

Make the password at least eight characters, if possible. A five-character password can be cracked by a computer program in less than one minute. A ten-character password, in contrast, has about 3,700 trillion possible character permutations and is considerably safer.

Choose passwords that are not in a dictionary—for instance, mix numbers and special characters with abbreviations or unusual words you will remember, but that do not conform to a pattern a computer can readily figure out.

Do not use your name, your kids' or pets' names, your address, your birthdate, or any other public information as your password.

Determine a *passphrase* that you can remember and use corresponding letters and symbols (such as the first letter of each word) for your password. For instance, the passphrase "My son John is five years older than my daughter Abby" could be used to remember the corresponding strong password "Msji5yotMd@".

Do not keep a written copy of the password in your desk or taped to your monitor. If you need to write down your password, create a password-protected file on your computer that contains all your passwords so you can look them up as needed.

Use a different password for your highly sensitive activities (such as online banking or stock trading) than for Web sites that remember your settings or profile (such as online news, auction, shopping, or bookstore sites). If a hacker determines your password on a low-security site (which is easier to break into than a site located on a secure Web server), he or she can use it on an account containing sensitive data if you use the same password on both accounts.

Change your passwords frequently.

Possessed Knowledge Systems

- Two-factor authentication: Use two different factors for increased security
 - Possessed knowledge (something you know)
 - Possessed object (something you have)
 - Biometric (something you are)
- OTP tokens are one example

FIGURE 9-5

Two-factor authentication. With this system, the user must have both the access card (to obtain the OTP) and his or her conventional username/password combination, in order to log on to his or her online account.

1. Press the button to display the OTP passphrase.

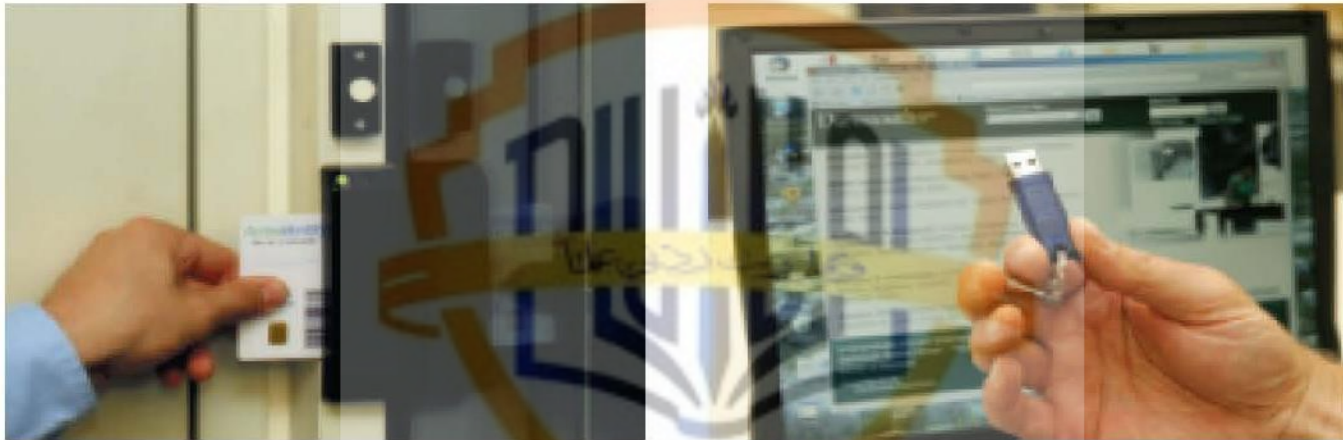
2. The displayed OTP passphrase is used in the PayPal login process.



Access Control Systems

- Possessed object access systems: Use a physical object an individual has in his/her possession to identify that individual
 - Smart cards
 - RFID-encoded badges
 - Magnetic cards
 - USB security keys or e-tokens
- Disadvantages: Can be lost or used by an unauthorized individual
 - When used with passwords or biometrics = two-factor authentication

Possessed Object Systems



SMART CARDS

Are read by a smart card reader to provide access to a facility or computer system.

USB SECURITY TOKENS

Are inserted into one of the computer's USB ports to provide access to that computer system.



FIGURE 9-6

Possessed objects.

Help protect against unauthorized access; some can also store additional security credentials.

Access Control Systems

- Biometric access systems: Identifies users by a particular unique biological characteristic
 - Fingerprint, hand, face, iris, voice, etc.
 - Data read by biometric reader must match what is stored in a database
- Often used:
 - To control access to secure facilities
 - To log on to computers, punch in/out at work, law enforcement, etc.
- Advantages: Can only be used by the authorized individual and cannot be lost or forgotten
- Disadvantages: Cannot be reset; expensive

Biometric Systems



FINGERPRINT RECOGNITION SYSTEMS

Typically used to protect access to office computers, to automatically supply Web site passwords on home computers, and to pay for products or services.



HAND GEOMETRY SYSTEMS

Typically used to control access to facilities (such as government offices, prisons, and military facilities) and to punch in and out of work.



FACE RECOGNITION SYSTEMS

Typically used to control access to highly secure areas, as well as to identify individuals for law enforcement purposes.



IRIS RECOGNITION SYSTEMS

Typically used to control access to highly secure areas and by the military; also beginning to be used to authenticate ATM users and other consumers.

FIGURE 9-7
Types of biometric access and identification systems.

Access Control Systems

- Controlling access to wireless networks
 - In general, Wi-Fi is less secure than wired networks
 - Security is usually off by default; wireless networks should be secured
 - Wireless network owners should:
 - Enable encryption (WPA is more secure than WEP)
 - Not broadcast the network name (SSID)
 - Enable other security features as needed

Controlling Access to Wireless Networks

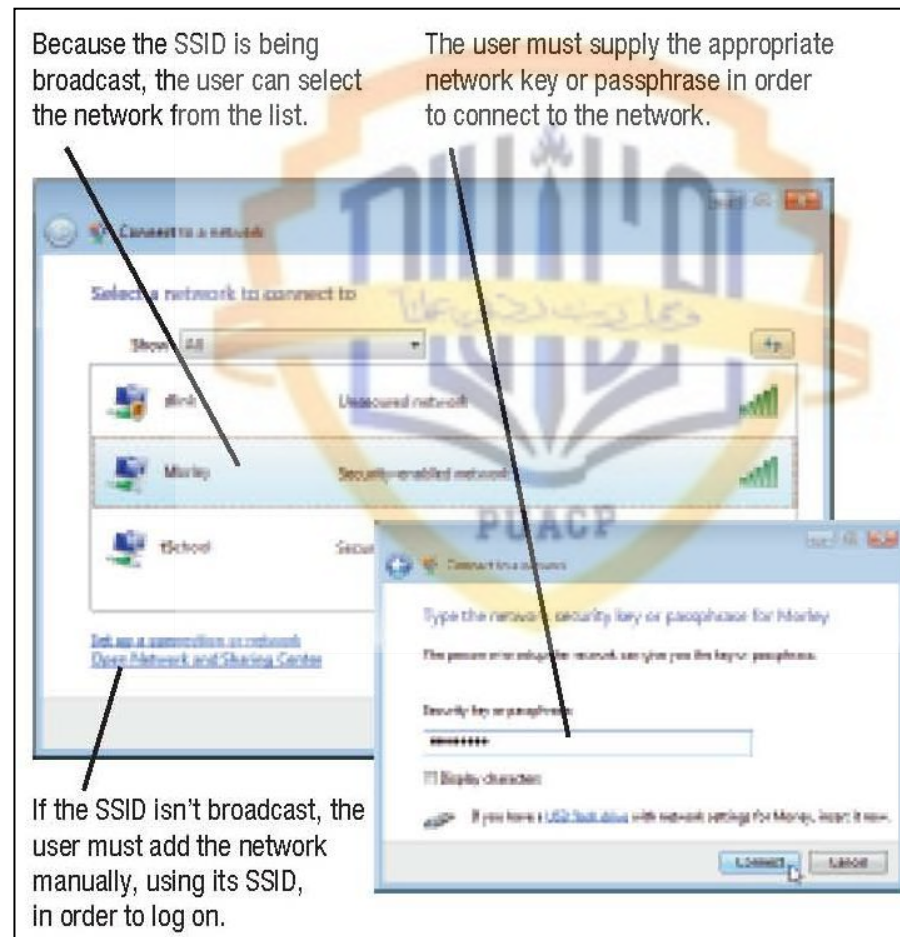
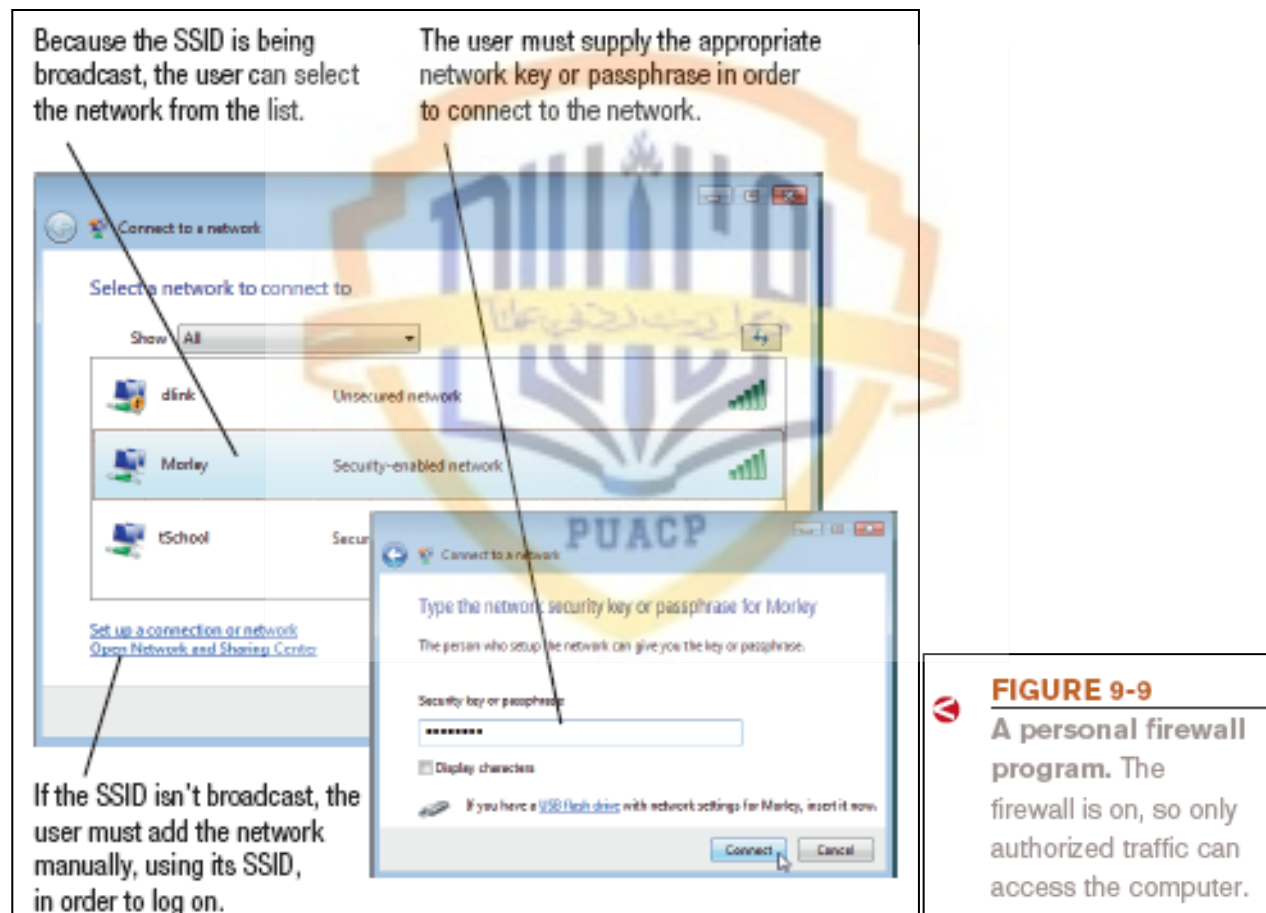


FIGURE 9-8
Accessing a Wi-Fi network. To access a secure network, the appropriate passphrase must be supplied.

Protecting Against Unauthorized Access and Use

- Firewall: A collection of hardware and/or software intended to protect a computer or computer network from unauthorized access
 - Blocks access to the computer from hackers
 - Blocks access to the Internet from programs on the user's computer unless authorized by the user
 - Important for home computer that have a direct Internet connection, as well as for businesses
 - Work by closing down external communications ports
- Intrusion prevention system (IPS) software: Monitors traffic to try and detect possible attacks

Firewalls



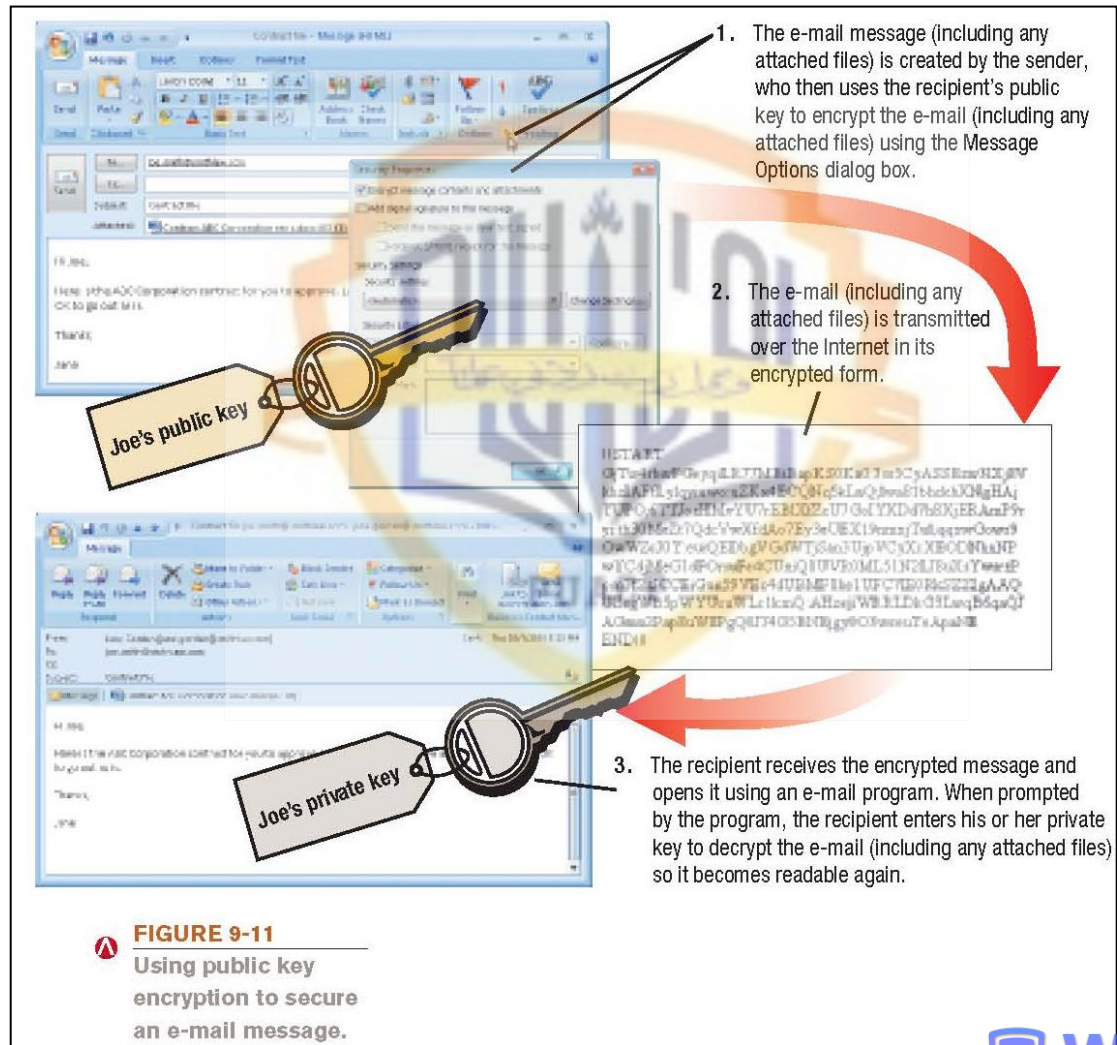
Protecting Against Unauthorized Access and Use

- Encryption: Method of scrambling contents of e-mail or files to make them unreadable if intercepted
 - Private key encryption: Uses a single key
 - Most often used to encrypt files on a computer
 - If used to send files to others, the recipient needs to be told the key
 - Public key encryption: Uses two keys
 - Public key: Can be given to anyone; used to encrypt messages to be sent to that person
 - Private key: Only known by the individual; used to decrypt messages that are encrypted with the individual's public key
 - Key pairs can be obtained through a Certificate Authority

Protecting Against Unauthorized Access, Use, and Computer Sabotage

- Secure Web pages: Use encryption (SSL, EV SSL, etc.) to protect information transmitted via their Web pages
 - Look for a locked padlock on the status bar and https:// in the URL
 - Only transmit credit card numbers and other sensitive data via a secure Web server
- Web-based encrypted e-mail (HushMail) is available
- Various strengths of encryption available
 - Stronger is more difficult to crack
 - Strong = 128-bit (16-character keys)
 - Military = 2,048-bit (256-character keys)

Encryption



Protecting Against Unauthorized Access and Use

- Virtual private networks (VPNs): A private secure path over the Internet
 - Allows authorized users to securely access a private network via the Internet
 - Much less expensive than a private secure network since uses the Internet
 - Can provide a secure environment over a large geographical area
 - Typically used by businesses to remotely access corporate networks via the Internet
 - Personal VPNs can be used by individuals to surf safely at a wireless hotspot

Protecting Against Unauthorized Access and Use

- Individuals should take additional precautions when using public hotspots in addition to using security software, secure Web pages, VPNs, and file encryption
 - Turn off file sharing
 - Disable Wi-Fi and Bluetooth if not needed
 - Use firewall to block incoming connections
 - Turn off automatic and ad hoc connections

PUBLIC HOTSPOT PRECAUTIONS

Turn off automatic connections and pay attention to the list of available hotspots to try to make sure you connect to a legitimate access point (not an evil twin).

Use a personal firewall to control the traffic going to and from your computer and temporarily use it to block all incoming connections.

Use a virtual private network (VPN) to secure all activity between your computer and the Internet.

Only enter passwords, credit card numbers, and other data on secure Web pages using a VPN.

If you're not using a VPN, encrypt all sensitive files before transferring or e-mailing them.

If you're not using a VPN, avoid online shopping, banking, and other sensitive transactions.

Turn off file sharing so others can't access the files on your hard drive.

Turn off Bluetooth and Wi-Fi when you are not using them.

Disable *ad hoc* capabilities to prevent another computer from connecting to your computer directly without using an access point.

Use antivirus software and make sure your operating system is up to date.



FIGURE 9-12

Sensible precautions for public Wi-Fi hotspot users.

Protecting Against Unauthorized Access and Use

- Sensible employee precautions
 - Screen potential new hires carefully
 - Watch for disgruntled employees and ex-employees
 - Develop policies and controls
 - Use data-leakage prevention and enterprise rights-management software
 - Ask business partners to review their security to avoid attacks coming from someone located at that organization

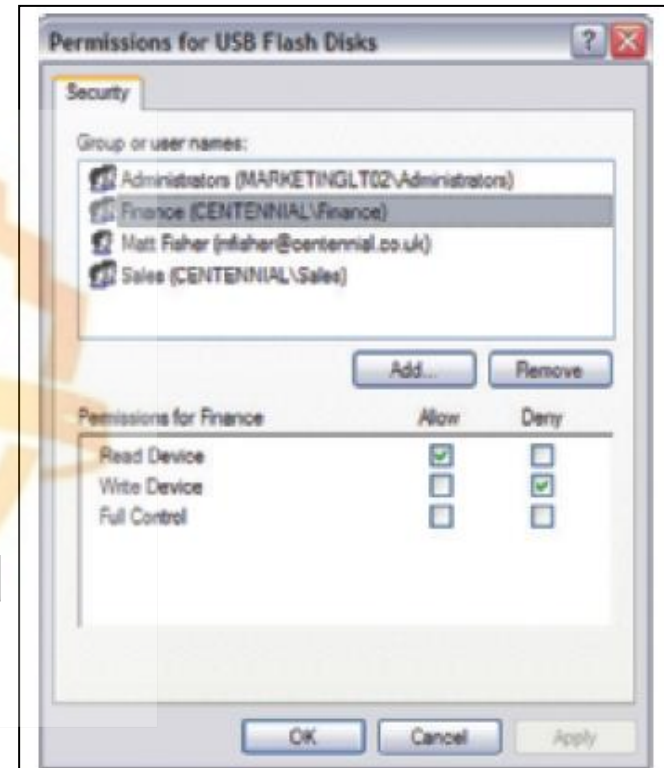


FIGURE 9-13
Data-leakage prevention software can control which devices can be connected to an employee's computer.

Quick Quiz

1. Which of the following is an example of possessed knowledge?
 - a. Password
 - b. Smart card
 - c. Fingerprint
2. True or False: With public key encryption, a single key is used to both encrypt and decrypt the file.
3. A(n) _____ controls access to a computer from the Internet and protects programs installed on a computer from accessing the Internet without authorization from the user.

Answers:

1) a; 2) False; 3) firewall

Computer Sabotage

- Computer sabotage: Acts of malicious destruction to a computer or computer resource
- Botnet: A group of bots (computers controlled by a criminal) that are controlled by one individual
 - Used by botherders to send spam, launch Internet attacks and malware, etc.
- Malware: Any type of malicious software
 - Written to perform destructive acts (damaging programs, deleting files, erasing drives, etc.)
 - Writing malware is considered unethical, distributing is illegal
 - Can infect mobile phones and mobile devices (some preinstalled on mobile devices)

Types of Malware

- Computer virus: A software program installed without the user's knowledge and designed to alter the way a computer operates or to cause harm to the computer system
 - Often embedded in downloaded programs and e-mail messages (games, videos, music files)
- Computer worm: Malicious program designed to spread rapidly by sending copies of itself to other computers
 - Typically sent via e-mail

Malware

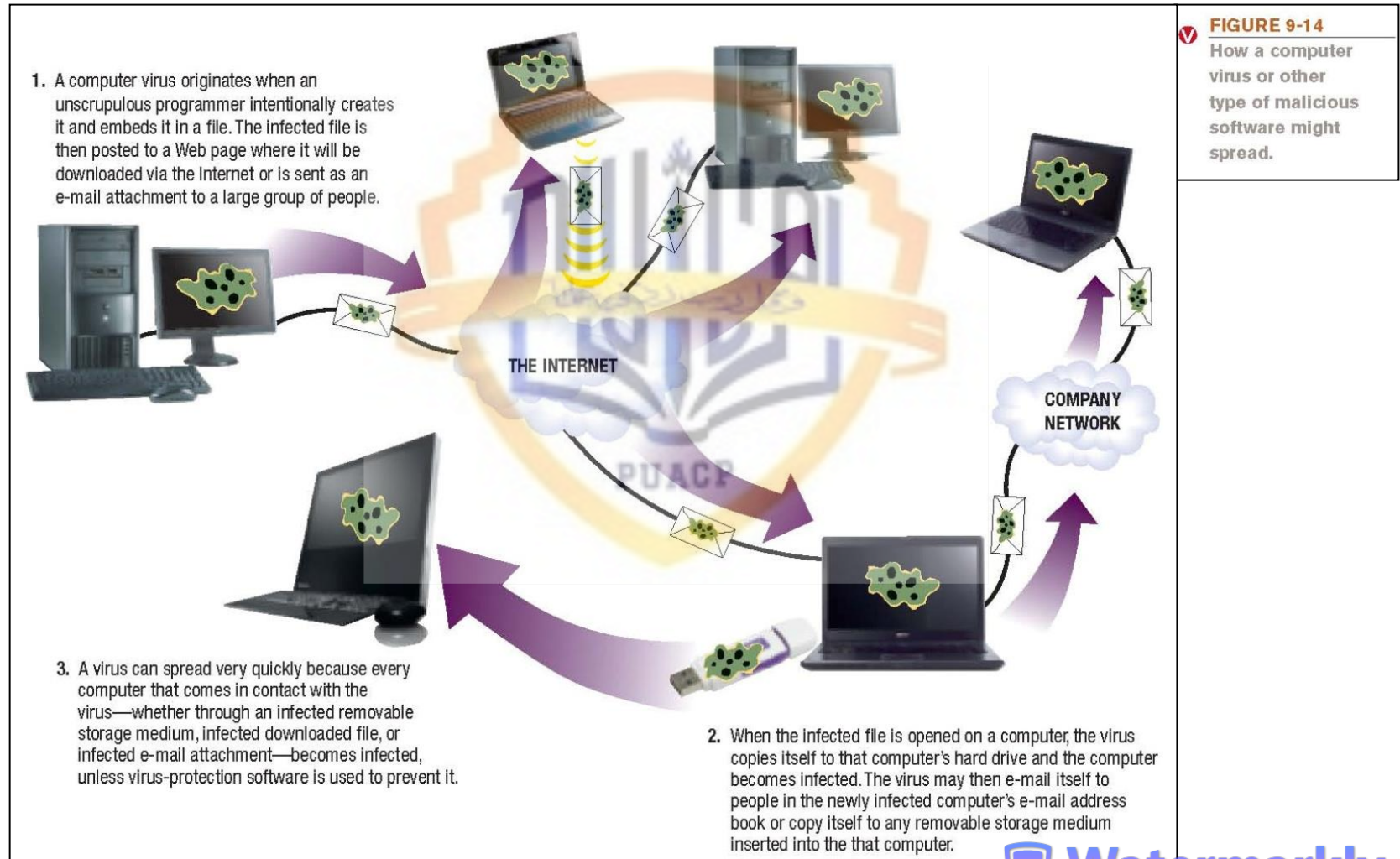


FIGURE 9-14
How a computer virus or other type of malicious software might spread.

Types of Malware

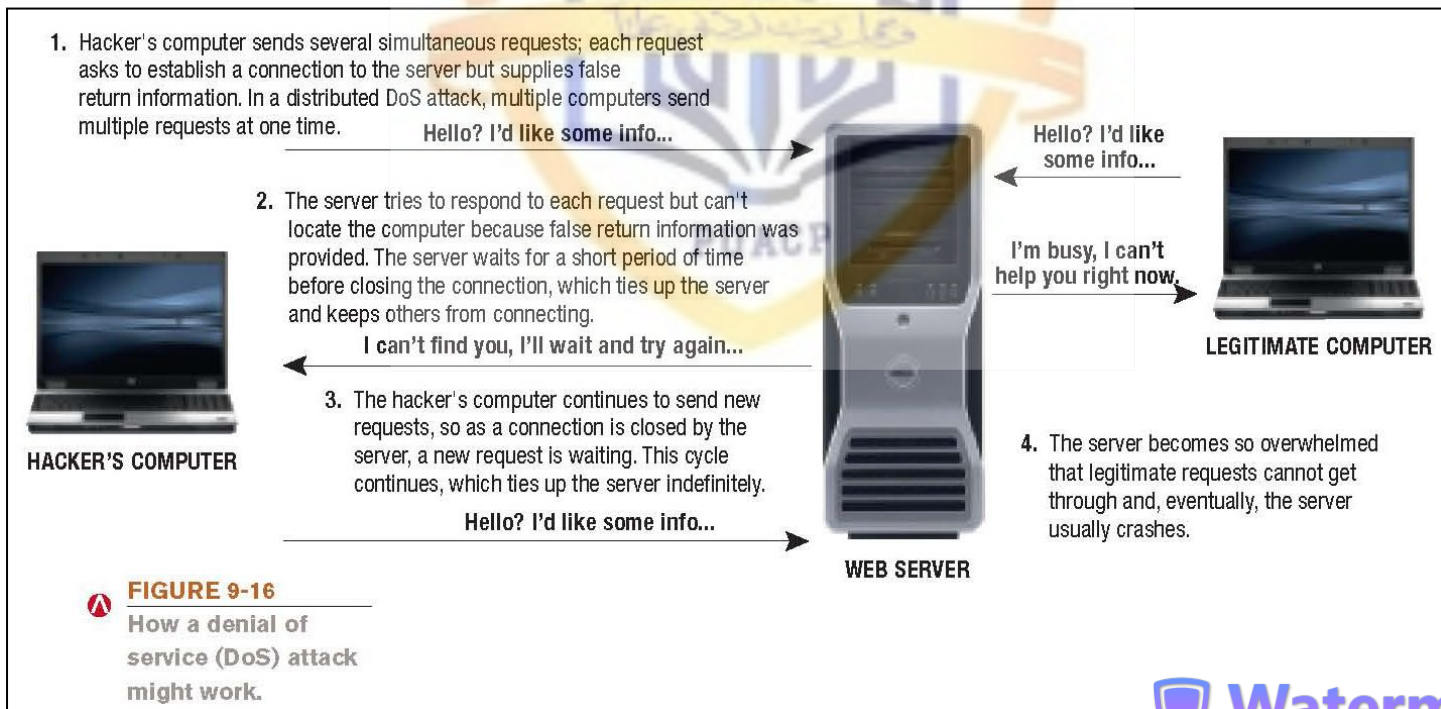
- Trojan horse: Malicious program that masquerades as something else
 - Usually appear to be a game or other program
 - Cannot replicate themselves; must be downloaded and installed
 - Rogue antivirus programs are common today
- Mobile malware: Becoming more common

FIGURE 9-15
Rogue antivirus programs. These programs try to trick victims into purchasing subscriptions to remove nonexistent malware supposedly installed on their computers.



Computer Sabotage

- Denial of service (DoS) attack: Act of sabotage that attempts to flood a network server or Web server with so much activity that it is unable to function
 - Distributed DoS attack: Uses multiple computers



Computer Sabotage

- Data or program alteration: When a hacker breaches a computer system in order to delete or change data
 - Students changing grades
 - Employees performing vengeful acts, such as deleting or changing corporate data
- Web site alteration: Changing content of a Web site
 - Web sites defaced to make political statements
 - Hacking into and changing social networking account contents (Facebook pages, Twitter tweets, etc.)
 - Altering legitimate site to perform malware attacks

Protecting Against Computer Sabotage

- Security software: Typically a suite of programs, used to protect your computer against a variety of threats
- Antivirus software: Used to detect and eliminate computer viruses and other types of malware
 - Should be set up to run continuously to check incoming e-mail messages, instant messages, Web page content, and downloaded files
 - Quarantines any suspicious content as it arrives
 - Regular system scans should be performed
 - New malware is introduced at all times, best to automatically download new virus definitions on a regular basis

Protecting Against Computer Sabotage

ANTIVIRUS SOFTWARE

Security History - Advanced Details

Alert Summary

Category	Entity	Date & Time	Status	Recommendation
Info	Quick Scan Results	11/10/2008 12:27:30 AM	Completed	

Advanced Details

Task Name	Quick Scan
Scan Time	11/10/2008 12:27:30 AM
Total Items Scanned	11,324
Total Security Risks Detected	1
Total Security Risks Pending	1
Total Security Risks Resolved	0

Norton 360

Email Scanner detected Trojan Horse in your email. The security risk was removed.

[View Details](#)

If malware is found during a scan or as you use your computer, the software removes it.

Ad-Aware Pro Standard Edition

Scan Results

Scan Mode: Custom manual scan
Scan Time: 00:00:30
Objects Scanned: 12417
Objects Detected: 76

Family	Category	Quantity	TAE	Action
Win32TrjUserBage	Malware	1	10	Recommended
Win32TrjUserBage is the download element from the Bage worm family. It downloads files and executes them on the victim's PC.				
File	C:\Documents and Settings\user\14367816766b01094c6e1			Quarantine
Win32TrjUserBage	Malware	1	10	Recommended
Win32TrjUserBage	Malware	1	10	Recommended
Win32TrjUserBage	Malware	1	10	Recommended
Win32TrjUserBage	Malware	1	10	Recommended
Win32TrjUserBage	Malware	1	10	Recommended

☐ Set System Restore Point

[Perform Actions Now](#)

Why does Lavasoft recommend this action?

ANTISPYWARE SOFTWARE

Both programs typically monitor your system on a continual basis, as well as periodically scanning your entire computer.

If spyware is found, the software recommends quarantining or removing it.

FIGURE 9-17
Security software.
Most security software is set up to monitor your system on a continual basis, removing threats as they are discovered.

Protecting Against Computer Sabotage

- Some ISPs filter include virus checking
- E-mail authentication systems can protect against viruses sent via e-mail
- Common sense precautions can help prevent a virus infection
- Web browser security settings can help protect against some attacks

VIRUS PREVENTION STRATEGIES

Use antivirus software to check incoming e-mail messages and files, and download updated virus definitions on a regular basis.

Limit the sharing of flash memory cards, USB flash drives, and other removable storage media with others.

Only download files from reputable sites.

Only open e-mail attachments that come from people you know and that do not have an executable file extension (such as .exe, .com, .bat, or .vbs); double-check with the sender before opening an unexpected, but seemingly legitimate, attachment.

For any downloaded file you are unsure of, upload it to a Web site (such as VirusTotal.com) that tests files for viruses before you open them.

Keep the preview window of your e-mail program closed so you will not view messages until you determine that they are safe to view.

Regularly download and install the latest security patches available for your operating system, browser, and e-mail programs.

Avoid downloading files from P2P sites.



FIGURE 9-18

Sensible precautions can help protect against computer virus infections.

Quick Quiz

1. Which of the following is used to control your computer by someone else?
 - a. Worm
 - b. Trojan horse
 - c. Botnet
2. True or False: Computer viruses can only be spread via the Internet.
3. A(n) _____ is a type of malware that masquerades as something else

Answers:

1) c; 2) False; 3) Trojan horse

Online Theft, Online Fraud, and Other Dot Cons

- Dot con: A fraud or scam carried out through the Internet
- Data theft or information theft can be committed by:
 - Stealing an actual computer or mobile device
 - A hacker gaining unauthorized access
 - Includes personal data, proprietary corporate information, and money
- Identity theft: Using someone else's identity to purchase goods or services, obtain new credit cards or bank loans, or illegally masquerade as that individual
 - Information obtained via documents, phishing schemes, stolen information, etc.
 - Expensive and time consuming to recover from

Identity Theft

FIGURE 9-19
How identity theft works.

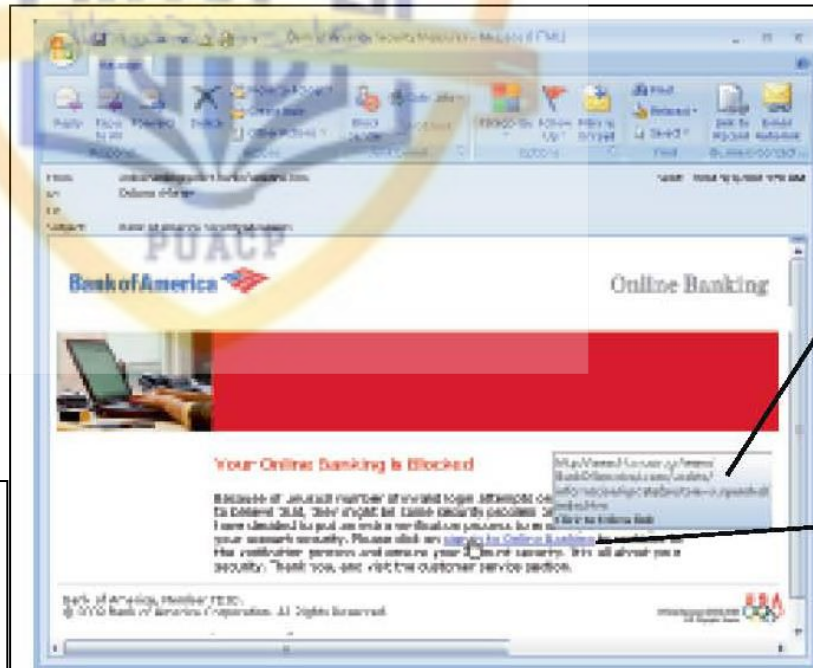


1. The thief obtains information about an individual from discarded mail, employee records, credit card transactions, Web server files, or some other method.
2. The thief uses the information to make purchases, open new credit card accounts, and more in the victim's name. Often, the thief changes the address on the account to delay the victim's discovery of the theft.
3. The victim usually finds out by being denied credit or by being contacted about overdue bills generated by the thief. Clearing one's name after identity theft is time-consuming and can be very difficult and frustrating for the victim.

Online Theft, Online Fraud, and Other Dot Cons

- Phishing: Use of spoofed e-mail messages to gain credit card numbers and other personal data
 - Typically contains a link to a spoofed Web site
 - After victim clicks a link in the message and supplies sensitive data, that data is sent to the thief
 - E-mails and Web sites often look legitimate

FIGURE 9-20
Phishing. Phishing schemes use legitimate-looking e-mails to trick users into providing private information.



The link is for an insecure Web page and does not use the bofa.com domain.

This e-mail looks legitimate, but the link goes to a spoofed Web page.

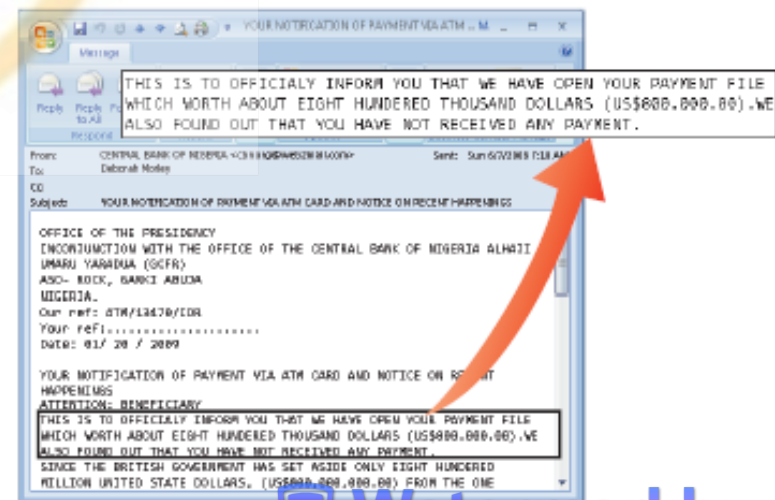
Online Theft, Online Fraud, and Other Dot Cons

- Spear phishing: A personalized phishing scheme targeted to specific individuals
 - Often include personalized information to seem more legitimate
 - May impersonate someone in your organization, such as from human resources or the IT dept.
- Pharming: The use of spoofed domain names to obtain personal information
 - DNS servers are hacked to route requests for legitimate Web pages to spoofed Web pages (DNS poisoning)
 - Often take place via company DNS servers
- Drive-by pharming: Hacker changes the DNS server used by a victim's router to use hacker's DNS server

Online Theft, Online Fraud, and Other Dot Cons

- Online auction fraud: When an item purchased through an online auction is never delivered, or the item is not as specified by the seller
- Internet offer scams: A wide range of scams offered through Web sites or unsolicited e-mails
 - Loan and pyramid scams
 - Work-at-home cons
 - Nigerian letter fraud scheme
 - Soliciting of donations after disasters
 - Pornographic sites
 - Fake job site postings

FIGURE 9-21
A Nigerian letter fraud e-mail.



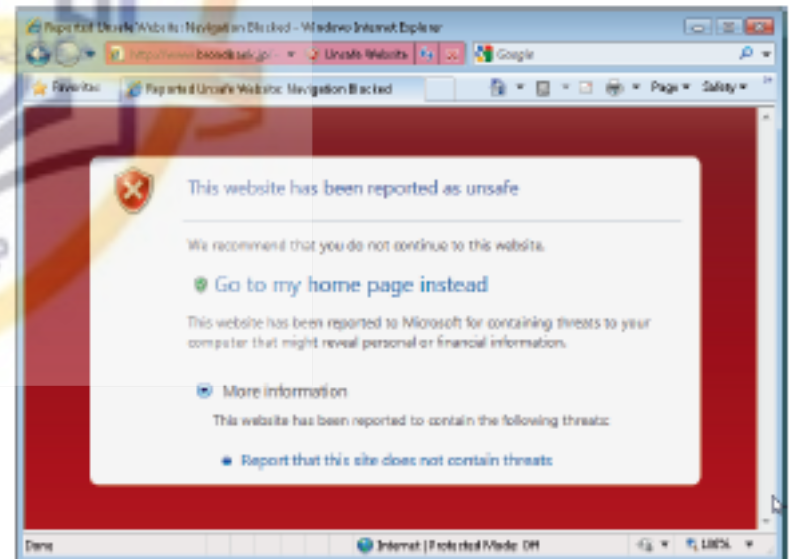
Protecting Against Online Theft, Online Fraud, and Other Dot Cons

- Protecting against data, information, and identity theft
 - Do not give out personal information (Social Security number, mother's maiden name, etc.) unless absolutely necessary
 - Never give out sensitive information over the phone or by e-mail
 - Shred documents containing sensitive data, credit card offers, etc.
 - Don't place sensitive outgoing mail in your mailbox
 - Watch your bills and credit report to detect identity theft early
 - Can get a free credit report from 3 major consumer credit bureaus each year

Protecting Against Online Theft, Online Fraud, and Other Dot Cons

- Protecting against phishing attacks
 - Never click a link in an e-mail to go to a secure Web site
 - Antiphishing tools built into Web browsers can help warn you of potential phishing sites
 - Some secure sites use methods to reassure users they are on the legitimate site

FIGURE 9-24
Unsafe Web site alerts.



Protecting Against Online Theft, Fraud, and Other Dot Cons

A PHISHING E-MAIL OFTEN . . .

Tries to scare you into responding by sounding urgent, including a warning that your account will be cancelled if you do not respond, or telling you that you have been a victim of fraud.

Asks you to provide personal information, such as your bank account number, an account password, credit card number, PIN number, mother's maiden name, or Social Security number.

Contains links that do not go where the link text says it will go (point to a hyperlink in the e-mail message to view the URL for that link).

Uses legitimate logos from the company the phisher is posing as.

Appears to come from a known organization, but one you may not have an association with.

Appears to be text or text and images but is actually a single image; it has been created that way to avoid being caught in a spam filter (a program that sorts e-mail based on legitimate e-mail and suspected spam) since spam filters cannot read text that is part of an image in an e-mail message.

Contains spelling or grammatical errors.



FIGURE 9-22

Tips for identifying phishing e-mail messages.

Protecting Against Online Theft, Fraud, and Other Dot Cons

TIPS FOR AVOIDING IDENTITY THEFT

Protect your Social Security number—give it out only when necessary.

Be careful with your physical mail and trash—shred all documents containing sensitive data.

Secure your computer—update your operating system and use up-to-date security (antivirus, antispymware, firewall, etc.) software.

Be cautious—never click on a link in an e-mail message or respond to a too-good-to-be-true offer.

Use strong passwords for your computer and online accounts.

Verify sources before sharing sensitive information—never respond to e-mail or phone requests for sensitive information.

Be vigilant while on the go—safeguard your wallet, mobile phone, and portable computer.

Watch your bills and monitor your credit reports—react immediately if you suspect fraudulent activity.

Use security software or browser features that warn you if you try to view a known phishing site.



FIGURE 9-23

Tips to reduce your risk of identity theft.

Protecting Against Online Theft, Online Fraud, and Other Dot Cons

- Digital certificate: Group of electronic data that can be used to verify the identity of a person or organization
 - Obtained from a Certificate Authority
 - Typically contains identity information about the person or organization, an expiration date, and a pair of keys to be used with encryption and digital signatures
 - Are also used with secure Web sites to guarantee that the site is secure and actually belongs to the stated individual or organization
 - Can be SSL or EV SSL

Protecting Against Online Theft, Online Fraud, and Other Dot Cons



Protecting Against Online Theft, Online Fraud, and Other Dot Cons

- Digital signature: Unique digital code that can be attached to an e-mail message or document
 - Can be used to verify the identity of the sender
 - Can be used to guarantee the message or file has not been changed
 - Uses public key encryption
 - Document is signed with the sender's private key
 - The key and the document create a unique digital signature
 - Signature is verified using the sender's public key

Protecting Against Online Theft, Online Fraud, and Other Dot Cons

- Protecting against other dot cons:
 - Use common sense
 - Check online auction seller's feedback before bidding
 - Pay for online purchases via a credit card so transactions can be disputed if needed
 - Use an online payment system
 - Take advantage of buyer protection
 - Use an escrow service for high-priced items

Personal Safety Issues

- Cyberbullying: Children or teenagers bullying other children or teenagers via the Internet
 - Common today, estimate 50% of all US teenagers
- Cyberstalking: Repeated threats or harassing behavior between adults carried out via e-mail or another Internet communication method
 - Sending harassing e-mail messages to the victim
 - Sending unwanted files to the victim
 - Posting inappropriate messages about the victim
 - Signing the victim up for offensive material
 - Publicizing the victim's contact information
 - Hacking into victim's social networking pages
- Sometimes escalates to personal violence  Watermarkly

Personal Safety Issues

- Online pornography
 - Concern for parents and schools
 - Difficult to stop due to constitutional rights
 - Online pornography involving minors is illegal
 - Link between online pornography and child molestation
 - Internet can make it easier to arrange dangerous meetings between predators and children

Protecting Against Cyberbullying and Cyberstalking

- Safety tips for adults
 - Be cautious in chat rooms and use gender-neutral, nonprovocative names
 - Do not reveal personal information
 - Do not respond to insults or harassing comments
 - Request to have personal information removed from online directories
- Safety tips for children
 - Parents should monitor Internet activities
 - Have children use a computer in a family room
 - They should be told which activities are allowed
 - Instruct them to tell a parent of a request for personal information or a personal meeting

Network and Internet Security Legislation

- It is difficult for the legal system to keep pace with the rate at which technology changes
- There are domestic and international jurisdictional issues
- Computer crime legislation continues to be proposed and computer crimes are being prosecuted

DATE LAW AND DESCRIPTION	
2004	Identity Theft Penalty Enhancement Act Adds extra years to prison sentences for criminals who use identity theft (including the use of stolen credit card numbers) to commit other crimes, including credit card fraud and terrorism.
2003	CAN-SPAM Act Implements regulations for unsolicited e-mail messages.
2003	Fair and Accurate Credit Transactions Act (FACTA) Amends the Fair Credit Reporting Act (FCRA) to require, among other things, that the three nationwide consumer reporting agencies (Equifax, Experian, and TransUnion) provide to consumers, upon request, a free copy of their credit report once every 12 months.
2003	PROTECT Act Includes provisions to prohibit virtual child pornography.

 **FIGURE 9-26**
Computer network and Internet security legislation.

Network and Internet Security Legislation

2003	Health Insurance Portability and Accountability Act (HIPAA) Includes a Security Rule that sets minimum security standards to protect health information stored electronically.
2002	Homeland Security Act Includes provisions to combat cyberterrorism, including protecting ISPs against lawsuits from customers for revealing private information to law enforcement agencies.
2002	Sarbanes-Oxley Act Requires archiving a variety of electronic records and protecting the integrity of corporate financial data.
2001	USA PATRIOT Act Grants federal authorities expanded surveillance and intelligence-gathering powers, such as broadening the ability of federal agents to obtain the real identity of Internet users, intercept e-mail and other types of Internet communications, follow online activity of suspects, expand their wiretapping authority, and more.
1998	Identity Theft and Assumption Deterrence Act of 1998 Makes it a federal crime to knowingly use someone else's means of identification, such as name, Social Security number, or credit card, to commit any unlawful activity.
1997	No Electronic Theft (NET) Act Expands computer piracy laws to include distribution of copyrighted materials over the Internet.
1996	National Information Infrastructure Protection Act Amends the Computer Fraud and Abuse Act of 1984 to punish information theft crossing state lines and to crack down on network trespassing.
1994	Computer Abuse Amendments Act Amends the Computer Fraud and Abuse Act of 1984 to include computer viruses and other harmful code.
1986	Computer Fraud and Abuse Act of 1986 Amends the 1984 law to include federally regulated financial institutions.
1984	Computer Fraud and Abuse Act of 1984 Makes it a crime to break into computers owned by the federal government. This act has been regularly amended over the years as technology has changed.

 **FIGURE 9-26**
Computer network
and Internet security
legislation.

Quick Quiz

1. Sending an e-mail that looks like it came from someone else in order to obtain information for fraudulent purposes is called _____.
 - a. hacking
 - b. online auction fraud
 - c. phishing
2. True or False: Cyberstalkers often find their victims online.
3. Using someone else's identity to purchase goods or services or perform other transactions is called _____.

Answers:

1) c; 2) True; 3) identity theft

Summary

- Why Be Concerned about Network and Internet Security?
- Unauthorized Access and Unauthorized Use
- Protecting Against Unauthorized Access & Unauthorized Use
- Computer Sabotage
- Protecting Against Computer Sabotage
- Online Theft, Online Fraud, and Other Dot Cons
- Protecting Against Online Theft, Online Fraud, and Other Dot Cons
- Personal Safety Issues
- Protecting Against Cyberbullying, Cyberstalking, and Other Personal Safety Concerns
- Network and Internet Security Legislation